



Cluster Analysis: Basic Concepts and Methods

IS-212
(Data and Knowledge Mining)

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References

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- Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Addison Wesley, 2006.
- Data Mining – Practical Machine Learning Tools and Techniques (2nd Ed.), Ian H. Witten and Eibe Frank, Morgan Kaufmann, 2005.
- Predictive Data Mining, S.M. Weiss and N. Indurkha, Morgan Kaufmann, 1998.
- Data Mining – Introductory and Advanced Topics, Margaret H. Dunham, Prentice Hall, 2003.
- Data Mining with R – Learning with Case Studies, Torgo, Luís, CRC Press, Taylor and Francis Group, LLC, 2011



Course Objectives

- To explain the workflow of data mining techniques
- To remember how to manipulate data mining techniques with Python and R-Programming
- To explain how to evaluate the experimental results of data mining techniques
- To express the visualization of the experimental results with Python and R-Programming



Outlines

- Cluster Analysis
- Partitioning Methods
- Hierarchical Methods
- Evaluation of Clustering
- Summary



Introduction

- **Clustering** is the process of grouping a set of data objects into multiple groups or clusters so that **objects within a cluster have high similarity**, but are **very dissimilar to objects in other clusters**.
- Cluster analysis or simply clustering is the process of partitioning a set of data objects (or observations) into subsets.
- Each subset is a cluster, such that objects in a cluster are similar to one another, yet dissimilar to objects in other clusters.
- The set of clusters resulting from a cluster analysis can be referred to as a clustering.
- Dissimilarities and similarities are assessed based on the attribute values describing the objects and often involve distance measures.
- Different clustering methods may generate different clusterings on the same data set.



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- Clustering is also called **data segmentation** in some applications because clustering partitions large data sets into groups according to their similarity.
- Clustering can also be used for **outlier detection**, where outliers (values that are “far away” from any cluster) may be more interesting than common cases.
- Clustering is known as **unsupervised** learning because the class label information is not present. For this reason, clustering is a form of **learning by observation**, rather than learning by examples.



Challenge

- Finding methods for efficient and effective cluster analysis in large databases.
 - **scalability** of clustering methods
 - the effectiveness of methods for clustering **complex shapes** (e.g., nonconvex) and **types of data** (e.g., text, graphs, and images)
 - **high-dimensional** clustering techniques (e.g., clustering objects with thousands of features)
 - methods for clustering mixed **numerical and nominal data** in large databases.



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- Orthogonal aspects with which clustering methods can be compared:
 - The partitioning criteria
 - Separation of clusters
 - Similarity measure
 - Clustering space - Subspace clustering discovers clusters and subspaces (often of low dimensionality) that manifest object similarity.



Review: Basic Cluster Analysis Methods

- Cluster Analysis: Basic Concepts
 - Group data so that object similarity is high within clusters but low across clusters
- Partitioning Methods
 - divides the data into k groups such that each group must contain at least one object
 - adopt exclusive cluster separation
 - Most partitioning methods are distance-based.
 - a good partitioning is that objects in the same cluster are “close” or related to each other, whereas objects in different clusters are “far apart” or very different.
 - K-means and k-medoids algorithms and their refinements
- Hierarchical Methods
 - Hierarchical clustering methods can be distance-based or density- and continuity based.
 - Hierarchical methods suffer from the fact that once a step (merge or split) is done, it can never be undone.
 - Agglomerative and divisive method



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- Density-based methods
 - Can find arbitrarily shaped clusters
 - continue growing a given cluster as long as the density (number of objects or data points) in the “neighborhood” exceeds some threshold.
 - May filter out outliers

- Grid-based methods
 - quantize the object space into a finite number of cells that form a grid structure.

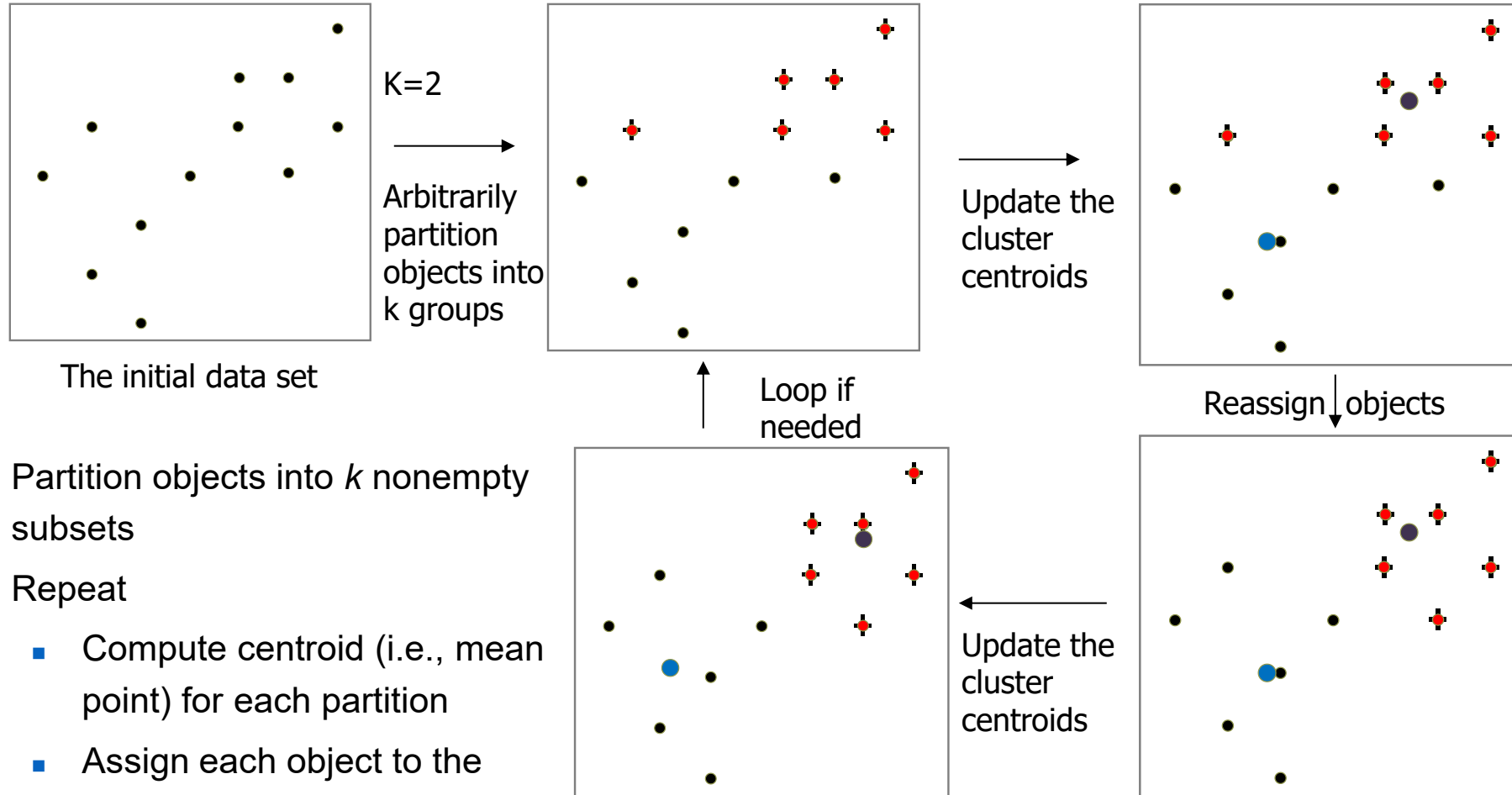


K-Means : a centroid-based technique

- A centroid-based partitioning technique uses the centroid of a cluster, C_i , as the representative of that cluster. (the centroid of a cluster is its center point)
- The centroid can be defined in various ways such as by the mean or medoid of the objects (or points) assigned to the cluster.
- The difference between an object $p \in C_i$ and c_i , the representative of the cluster, is measured by $\text{dist}(p, c_i)$, where $\text{dist}(x, y)$ is the Euclidean distance between two points x and y .
- The quality of cluster C_i can be measured by the **within-cluster variation**, which is the sum of squared error between all objects in C_i and the centroid c_i , defined as

$$E = \sum_{i=1}^k \sum_{p \in C_i} \text{dist}(p, c_i)^2$$

K-Means Clustering



- Partition objects into k nonempty subsets
- Repeat
 - Compute centroid (i.e., mean point) for each partition
 - Assign each object to the cluster of its nearest centroid
- Until no change



k-Medoids: a representative object-based technique

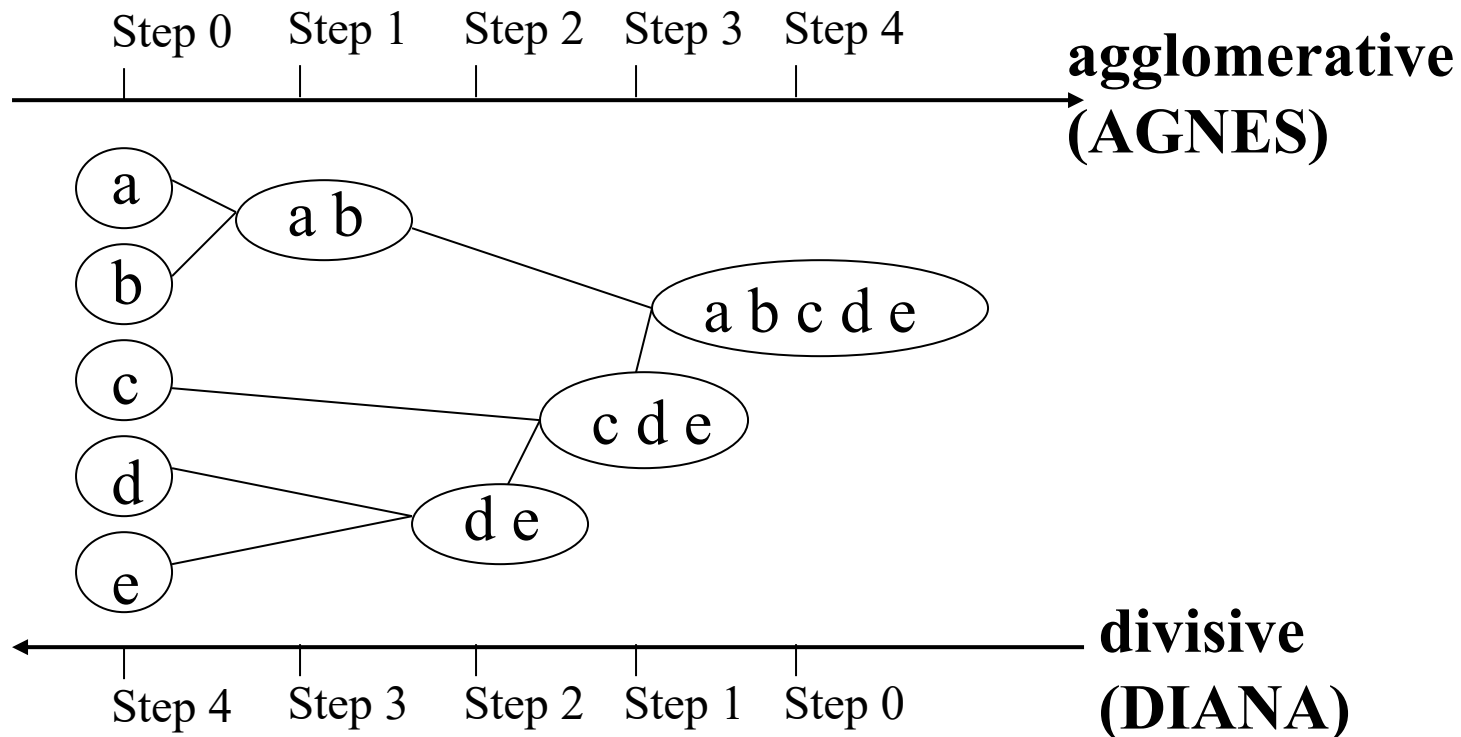
- The k-means algorithm is sensitive to outliers because such objects are far away from the majority of the data, and thus, when assigned to a cluster, they can dramatically distort the mean value of the cluster.
- Instead of taking the mean value of the objects in a cluster as a reference point, it can be picked actual objects to represent the clusters, using one representative object per cluster.
- Each remaining object is assigned to the cluster of which the representative object is the most similar.
- The partitioning method is then performed based on the principle of minimizing the sum of the dissimilarities between each object p and its corresponding representative object.
- An absolute-error criterion is used, defined as

$$E = \sum_{i=1}^k \sum_{p \in C_i} \text{dist}(p, o_i)$$



Hierarchical Clustering

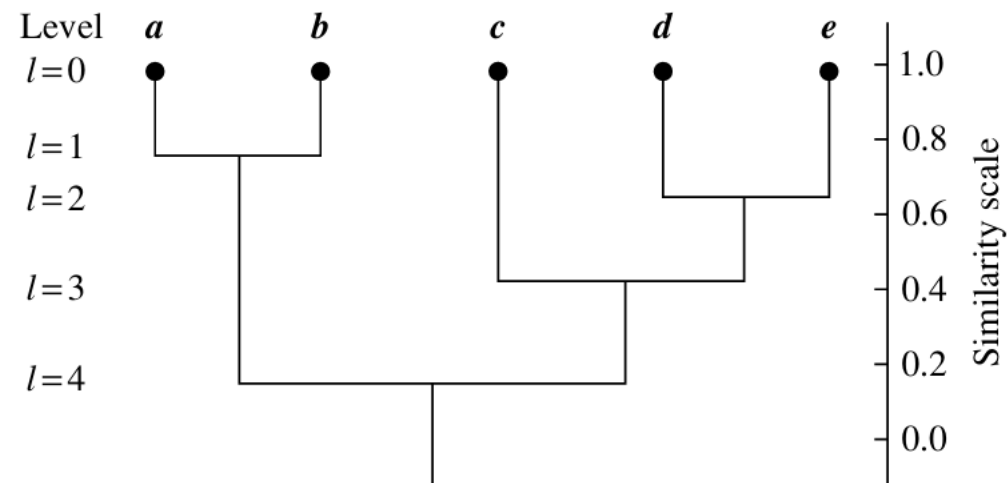
- Use distance matrix as clustering criteria. This method does not require the number of clusters k as an input, but needs a termination condition





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- A tree structure called a dendrogram is commonly used to represent the process of hierarchical clustering.
- It shows how objects are grouped together (in an agglomerative method) or partitioned (in a divisive method) step-by-step.



Dendrogram representation for hierarchical clustering of data objects $\{a, b, c, d, e\}$.



Distance between Clusters

- **Single link:** smallest distance between an element in one cluster and an element in the other, i.e., $\text{dist}(K_i, K_j) = \min(t_{ip}, t_{jq})$
- **Complete link:** largest distance between an element in one cluster and an element in the other, i.e., $\text{dist}(K_i, K_j) = \max(t_{ip}, t_{jq})$
- **Average:** avg distance between an element in one cluster and an element in the other, i.e., $\text{dist}(K_i, K_j) = \text{avg}(t_{ip}, t_{jq})$
- **Centroid:** distance between the centroids of two clusters, i.e., $\text{dist}(K_i, K_j) = \text{dist}(C_i, C_j)$
- **Medoid:** distance between the medoids of two clusters, i.e., $\text{dist}(K_i, K_j) = \text{dist}(M_i, M_j)$
 - Medoid: a chosen, centrally located object in the cluster



Evaluation of Clustering

- Assessing clustering tendency
- Determining the number of clusters in a data set
- Measuring clustering quality
 - extrinsic methods
 - intrinsic methods
 - The silhouette coefficient considers the average situations.

$$a(o) = \frac{\sum_{o' \in C_i, o \neq o'} \text{dist}(o, o')}{|C_i| - 1} \quad b(o) = \min_{C_j: 1 \leq j \leq k, j \neq i} \left\{ \frac{\sum_{o' \in C_j} \text{dist}(o, o')}{|C_j|} \right\}$$

- The silhouette coefficient of o is then defined as: $s(o) = \frac{b(o) - a(o)}{\max\{a(o), b(o)\}}$
- The value of the silhouette coefficient is between -1 and 1 .

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Determining the number of clusters

- Finding a good balance between compressibility and accuracy in cluster analysis
- The desired number of k is often dependent on the shape and scale of the distribution of points in a data set and the desired clustering resolution of the user.
- A simple method is to set the number of clusters to about $\sqrt{\frac{n}{2}}$ for a data set of n points.
In expectation, each cluster has $\sqrt{2n}$ points.
- The number of clusters k can be estimated by maximizing the Calinski-Harabasz index.

$$CH(k) = \frac{\frac{B(k)}{k-1}}{\frac{W(k)}{n-k}}$$

- The elbow method is based on the observation that increasing the number of clusters can help to reduce the sum of within-cluster variance of each cluster.



End of Lecture!

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